B. M. S. COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to VTU, Belagavi

DEPARTMENT OF MACHINE LEARNING

Program: B.E. In Artificial Intelligence and Machine Learning

Academic Year: 2026 - 27, Session: July 2026 - Dec 2026

Social Network Analysis (24AM7PESNA)

Alternative Assessment Tool (AAT) Report

Topic: Link Prediction in a Co-authorship Network: Recommending Future Research Collaborations

Submitted by

Student Name: Team Member 1: Team Member 2:	 Thijesh Praveen V Uday Kumar M
USN: Team Member 1: Team Member 2:	 1BM23AI200 1BM23AI203
Semester & Section:	

Valuation Report

(to be filled by faculty)

Score:	
Comments:	
Faculty In-charge:	Lavanya B Koppal
Faculty Signature with date:	

Table of Contents

1. Introduction . . . 3
- 1.1 Background of Social Network Analysis . . . 3
- 1.2 Importance of Link Prediction . . . 3
- 1.3 Objective of the Study . . . 3
- 1.4 Scope of the Report . . . 3

2. Problem Definition and Network Modeling . . . 4
- 2.1 Description of the Link Prediction Task . . . 4
- 2.2 Real-World Use Cases . . . 4
- 2.3 Relevance to the Syllabus and Industry . . . 4
- 2.4 Key Challenges . . . 4
- 2.5 Network Model . . . 4

3. Dataset and Network Statistics . . . 4
- 3.1 Dataset Description . . . 4
- 3.2 Data Cleaning Steps . . . 5
- 3.3 Network Statistics . . . 5
- 3.4 Central Authors . . . 6
- 3.5 Community Structure . . . 6

4. Methodology: Link Prediction Heuristics . . . 7
- 4.1 Theoretical Foundations . . . 7
- 4.2 Why Adamic-Adar Uses 1 / log k(w) . . . 7
- 4.3 Why These Heuristics Suit the Task . . . 7
- 4.4 Workflow . . . 7

5. Experimental Setup . . . 8
- 5.1 Train/Test Split . . . 8
- 5.2 Negative Sampling and Leakage Prevention . . . 8
- 5.3 Evaluation Metrics . . . 8
- 5.4 Tools, Libraries and Configuration . . . 8

6. Results and Evaluation . . . 8
- 6.1 Comparison of the Four Heuristics . . . 8
- 6.2 ROC and Precision-Recall Curves . . . 9
- 6.3 Precision@K and a Harder Test with Hard Negatives . . . 10
- 6.4 Stability over Five Random Splits . . . 10
- 6.5 Comparison and Analysis: Why Adamic-Adar Wins . . . 11

7. Recommendations . . . 11
- 7.1 Candidate Generation and Filters . . . 11
- 7.2 Validation: Do the Predictions Come True? . . . 11
- 7.3 Top-20 Recommended Collaborations (Adamic-Adar) . . . 12
- 7.4 Top-10 of the Other Methods . . . 13

8. Discussion . . . 13
- 8.1 Interpretation of Results . . . 13
- 8.2 Strengths and Weaknesses . . . 14
- 8.3 Limitations and Threats to Validity . . . 14
- 8.4 Challenges Encountered . . . 14

9. Conclusion . . . 14
- 9.1 Summary of the Work Done . . . 14
- 9.2 Key Learnings and Insights . . . 14
- 9.3 Real-World Applicability . . . 14
- 9.4 Future Improvements . . . 15

10. References . . . 15

Abstract

Co-authorship networks record who has written papers with whom. This report treats the recommendation of future research collaborations as a *link prediction* problem on the SNAP *ca-GrQc* network (5,242 authors, 14,484 co-authorship links after cleaning). On the largest connected component (4,158 authors, 13,422 links) we hide 2,684 links (20 %), using a spanning-tree-preserving split, and ask four classical neighbourhood heuristics - Common Neighbours, Jaccard, Adamic-Adar and Preferential Attachment - to rank them against an equal number of sampled non-links. The three local indices reach a ROC-AUC of 0.955-0.956, against 0.849 for Preferential Attachment. Adamic-Adar is the best method (mean ROC-AUC 0.9568 ± 0.0028 over five random splits), and its advantage becomes clear on a harder test with negatives that already share a co-author (ROC-AUC 0.850 vs 0.778 for Common Neighbours). Ranking all 46,588 distance-2 candidate pairs, 100% of its Top-100 predictions are real held-out collaborations (random baseline 5.3%). We finally produce Top-20 explained recommendations, visualisations and a Cytoscape-ready export.

Keywords: social network analysis, link prediction, co-authorship network, Adamic-Adar, collaboration recommendation, triadic closure.

1. Introduction

1.1 Background of Social Network Analysis

Social Network Analysis (SNA) models a system as a graph of *actors* (nodes) and *relationships* (edges) and studies the structure of these relationships: who is central, which groups form, how information flows and how the network evolves. Scientific collaboration is a classic SNA domain - Newman's studies of co-authorship networks showed that they are small worlds with high clustering and heavy-tailed degree distributions [15].

1.2 Importance of Link Prediction

Networks are not static. *Link prediction* asks which currently unconnected pairs of nodes are likely to become connected [1]. In a co-authorship network a predicted link is a **recommended collaboration**: two researchers who have not written together yet, but whose position in the network suggests that they should know each other.

1.3 Objective of the Study

- Model the ca-GrQc collaboration data as a graph and describe its structure.
- Implement four neighbourhood-based link predictors from scratch and verify them against NetworkX.
- Evaluate them without information leakage (ROC-AUC, Average Precision, Precision@K, P/R/F1) and check stability over several random splits.
- Produce explained Top-K collaboration recommendations and visualisations, including a Cytoscape export.

1.4 Scope of the Report

The study is limited to unsupervised, structure-only heuristics on a static, unweighted snapshot. Supervised and embedding-based methods are discussed as future work (Section 9.4).

2. Problem Definition and Network Modeling

2.1 Description of the Link Prediction Task

Let G = (V, E) be an undirected graph. A fraction of E is hidden as the test set E_{test}; the remaining training graph G_{train} = (V, E \ E_{test}) is the only information available to the predictor. A heuristic assigns a score s(u, v) to every non-adjacent pair of G_{train}; a good heuristic ranks the hidden pairs of E_{test} above pairs that never link.

2.2 Real-World Use Cases

- Collaborator recommendation in academic platforms (Google Scholar, ResearchGate) and "people you may know" features in professional networks.
- Team formation for grants and interdisciplinary projects; reviewer suggestion for journals.
- Forecasting how a research field will grow, and detecting missing or mis-attributed records in bibliographic databases.

2.3 Relevance to the Syllabus and Industry

The project uses the core SNA toolkit - graph modeling, degree distributions, clustering, path lengths, centrality (degree, betweenness, PageRank), community detection (Louvain) and link prediction - and the same evaluation discipline (held-out data, negative sampling, ranking metrics) that industry recommender systems use.

2.4 Key Challenges

- **Extreme class imbalance:** the LCC has 8,628,981 non-adjacent pairs for only 13,422 edges (643 non-links per link).
- **Leakage:** scores must be computed without the hidden edges, or evaluation is meaningless.
- **Cold start:** an author with few collaborators gives the local heuristics little to work with.
- **No timestamps:** SNAP's ca-GrQc has no dates, so the 'future' is simulated by random hiding.

2.5 Network Model

Table 1: *Mapping of the collaboration data to a network.*

Element	Mapping
Node	An author of a paper submitted to the arXiv General Relativity and Quantum Cosmology (GR-QC) category
Edge	The two authors co-authored at least one paper (Jan 1993 - Apr 2003)
Type	Undirected (co-authorship is symmetric), unweighted, no self-loops
Scope	Full graph for descriptive statistics; largest connected component (LCC) for all link-prediction experiments

A paper with k authors becomes a k-clique, so the network is built from overlapping cliques. This is why it is so strongly clustered, and why collaborators of collaborators so often become collaborators (*triadic closure*).

3. Dataset and Network Statistics

3.1 Dataset Description

The SNAP ca-GrQc dataset [2, 14] is a tab-separated edge list with 28,980 lines over 5,242 author IDs; SNAP stores every undirected pair in both directions. Author identities are anonymised integers.

3.2 Data Cleaning Steps

Table 2: *Cleaning log produced by* `data.clean_edges`.

Step	Result
Raw data lines read	28,980
Self-loop lines removed (author paired with themself)	12
Duplicate lines removed (reverse direction of each pair, after ordering u < v)	14,484
Nodes after cleaning (all IDs kept)	5,242
Undirected edges after cleaning	14,484
Authors isolated after cleaning (their only line was a self-loop)	1

3.3 Network Statistics

Table 3: *Basic network statistics (path statistics computed exactly on the LCC).*

Statistic	Full graph	LCC
Nodes	5,242	4,158
Edges	14,484	13,422
Density	0.001054	0.001553
Average degree	5.5261	6.4560
Maximum degree	81	81
Median degree	3	3
Average clustering coefficient	0.5296	0.5569
Transitivity (global clustering)	0.6298	0.6289
Connected components	355	1
Largest component size (nodes)	4,158	4,158
Isolated nodes	1	0
Degree assortativity	0.6593	0.6392
Diameter	-	17
Average shortest path length	-	6.0494

The network is very sparse (density 0.00105) yet strongly clustered: the average clustering coefficient (0.530) is about 502 times the density, i.e. far higher than in a random graph with the same density. It splits into 355 components, but the LCC holds 79.3% of the authors. Inside the LCC any two authors are on average 6.05 steps apart (diameter 17), a small-world structure. The positive degree assortativity (0.659) means that prolific authors mostly collaborate with other prolific authors, a consequence of large multi-author papers.

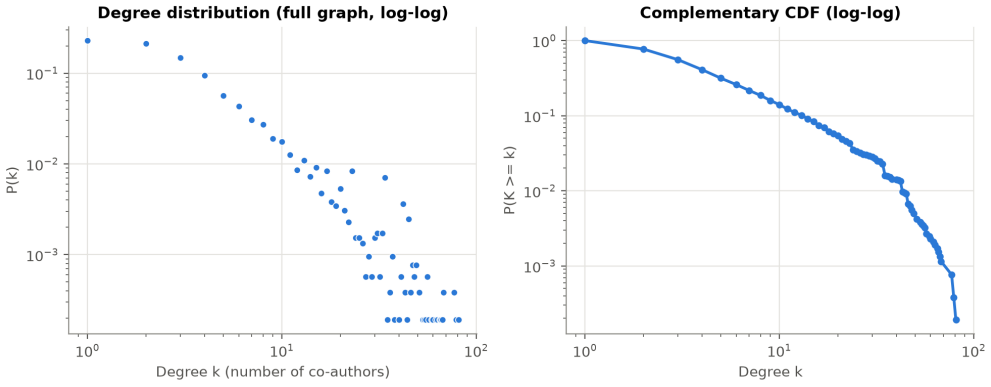


Figure 1: *Degree distribution of the full graph on log-log axes (left) and its complementary CDF (right). Median degree 3, maximum 81: a heavy tail of very collaborative authors.*

3.4 Central Authors

Table 4: *Top-10 authors by degree, exact betweenness and PageRank (full graph).*

#	Degree: author	k	Betweenness: author	value	PageRank: author	value
1	21012	81	13801	0.0370	14265	0.00145
2	21281	79	9572	0.0257	13801	0.00134
3	12365	77	14599	0.0255	13929	0.00131
4	22691	77	7689	0.0250	9572	0.00117
5	6610	68	13929	0.0247	2710	0.00115
6	9785	68	5052	0.0244	21281	0.00115
7	21508	67	14485	0.0235	7689	0.00110
8	17655	66	2710	0.0224	22691	0.00108
9	2741	65	14265	0.0197	6264	0.00107
10	19423	63	17655	0.0180	21012	0.00107

The rankings capture different roles. The highest-degree authors are members of very large multi-author collaborations; the highest-betweenness authors are *brokers* linking otherwise separate groups; PageRank rewards being connected to well-connected authors. No author appears in all three Top-10 lists, which shows how different these notions of importance are.

3.5 Community Structure

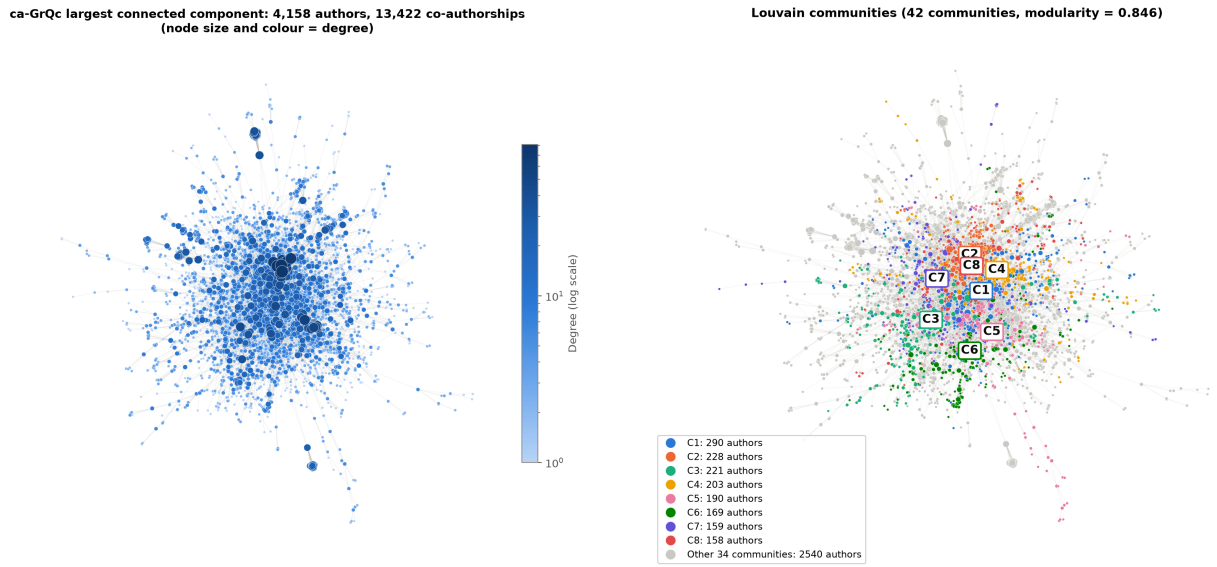


Figure 2: *Force-directed layout of the LCC. Left: node size and colour = degree. Right: Louvain communities [8] (42 communities, modularity 0.846); the eight largest are coloured and labelled, the rest grey.*

Louvain finds 42 communities with a very high modularity of 0.846; the largest holds only 7.0% of the LCC. Strong communities matter for link prediction: 88.6% of the held-out test links join two authors of the same community, so new collaborations are overwhelmingly local.

4. Methodology: Link Prediction Heuristics

4.1 Theoretical Foundations

Notation: Γ(x) is the set of co-authors (neighbours) of author x and k(x) = |Γ(x)| is its degree. All four heuristics are *unsupervised similarity scores*: a larger score means a link is more likely. They rest on two sociological mechanisms: **triadic closure** (people who share friends tend to become friends) and **preferential attachment** (well-connected people attract new connections) [5]. All four are implemented from scratch with Python sets in `src/scores.py`.

Table 5: *The four heuristics (Γ = neighbour set, k = degree).*

Heuristic	Formula	Intuition
Common Neighbours (CN)	$CN(u,v) = \lvert\Gamma(u) \cap \Gamma(v)\rvert$	Counts shared collaborators: the direct measure of triadic closure [4]. Simple and strong, but biased towards authors with many co-authors.
Jaccard Coefficient (JC)	$JC(u,v) = \frac{\lvert\Gamma(u) \cap \Gamma(v)\rvert}{\lvert\Gamma(u) \cup \Gamma(v)\rvert}$	Shared collaborators as a fraction of all collaborators of the pair [6]. Normalises CN so that prolific authors are not automatically favoured, but over-rewards pairs of authors with very few co-authors.
Adamic-Adar (AA)	$AA(u,v) = \sum_{w \in \Gamma(u) \cap \Gamma(v)} \frac{1}{\log k(w)}$	Like CN, but each shared collaborator w counts 1 / log k(w) [3]. A shared collaborator who works with few people is strong evidence; a hub who works with everyone is weak evidence.
Preferential Attachment (PA)	$PA(u,v) = k(u) \cdot k(v)$	Product of the two degrees [5]. Purely a popularity score: it ignores whether the two authors are close in the network.

4.2 Why Adamic-Adar Uses 1 / log k(w)

A collaborator w with degree k(w) 'introduces' k(w)(k(w)-1)/2 pairs of its co-authors to each other, so sharing a hub says little about a specific pair. Dividing by log k(w) down-weights hubs smoothly: a shared co-author with 3 collaborators contributes 1/log 3 = 0.91, one with 60 collaborators only 1/log 60 = 0.24 (natural log). The logarithm (rather than 1/k) keeps hubs from being ignored completely. Every shared neighbour has k(w) of at least 2, so log k(w) is always positive.

4.3 Why These Heuristics Suit the Task

Co-authorship links are produced by groups (papers) and grow by introductions through common collaborators, so neighbourhood overlap is a strong signal. The heuristics are interpretable (a recommendation can be explained by naming the shared collaborators), need no training, scale to the whole graph, and are the standard baselines of the link prediction literature [1, 7].

4.4 Workflow

```
load + clean edge list -> G (full) -> LCC
split LCC edges 80/20 (spanning tree always in train) -> G_train, test positives
sample non-edges of G (1:1) -> train / test negatives
for each pair: CN, JC, AA, PA computed on G_train only
threshold = argmax F1 on train pairs -> metrics on test pairs
repeat split for 5 seeds -> mean +/- std ROC-AUC
rank distance-2 candidates -> hit rate (G_train) and Top-K recommendations (full LCC)
```

5. Experimental Setup

5.1 Train/Test Split

Deleting 20 % of edges at random would cut bridges and isolate low-degree authors, who could then never be scored. We therefore first draw a random spanning tree of the LCC (4,157 edges, minimum spanning tree over random weights, seed 42). A spanning tree contains every bridge, so keeping it in the training graph guarantees that the training graph stays connected. The 2,684 test positives (20 % of 13,422) are then sampled uniformly from the remaining non-tree edges. The training graph keeps all 4,158 nodes, 10,738 edges and 1 connected component.

5.2 Negative Sampling and Leakage Prevention

Evaluation needs pairs that do *not* link. Negatives are sampled uniformly from pairs that are not edges of the **full** graph, so a hidden test edge can never be mislabelled as a negative. Sampling is 1:1: 10,738 train negatives and 2,684 test negatives, disjoint. Leakage is avoided by computing **every score on the training graph only**. Train positives are themselves edges of that graph, so they are scored with their own edge hidden. Otherwise PA and Jaccard would 'see' the edge they are meant to predict.

5.3 Evaluation Metrics

- **ROC-AUC**: probability that a random positive pair is scored above a random negative pair; threshold-free, uses the whole ranking.
- **Average Precision (AP)**: area under the precision-recall curve; sensitive to the top of the ranking.
- **Precision@K** (K = 10, 20, 50, 100): fraction of true links among the K highest-scored pairs - what a user of a recommender actually sees. Ties are broken randomly with seed 42.
- **Precision, Recall, F1 at a threshold**: the threshold maximising F1 on the *training* pairs is applied unchanged to the test pairs.
- **Hit rate@K**: fraction of the Top-K over *all* distance-2 candidate pairs of the training graph that are real held-out links.

5.4 Tools, Libraries and Configuration

Table 6: *Environment and configuration.*

Item	Value
Language / libraries	Python 3.14.5, NetworkX 3.7, pandas 3.1.0, NumPy 2.5.4, scikit-learn 1.9.1, Matplotlib 3.11.2, pyvis, ReportLab, python-pptx
Random seed	42 (split, sampling, tie-breaking, layouts, Louvain); stability seeds 42, 43, 44, 45, 46
Split / negatives	80 % train / 20 % test edges of the LCC; negatives 1:1, from non-edges of the full graph
Hard-negative test	same 2,684 positives + 2,684 non-links at distance 2 in the training graph
Implementation check	own scores vs NetworkX on 16,106 pairs: max absolute difference 0
Runtime	whole notebook 4.6 min on a laptop

6. Results and Evaluation

6.1 Comparison of the Four Heuristics

Table 7: *Test-set results (2,684 positives + 2,684 random negatives, seed 42). The threshold is the F1-optimal value learned on the training pairs.*

Method	ROC-AUC	AP	Thr.	Prec.	Recall	F1	P@10	P@20	P@50	P@100
Common Neighbours	0.9554	0.9550	1	0.995	0.912	0.952	1.000	1.000	1.000	1.000
Jaccard	0.9555	0.9555	0.0119	0.995	0.912	0.952	1.000	1.000	1.000	1.000
Adamic-Adar	0.9557	0.9558	0.2423	0.995	0.912	0.952	1.000	1.000	1.000	1.000
Preferential Attachment	0.8490	0.8662	10	0.652	0.902	0.756	1.000	1.000	1.000	1.000

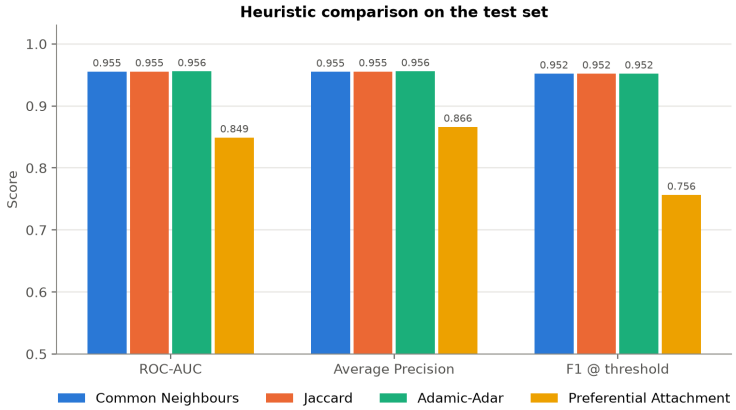


Figure 3: *ROC-AUC, Average Precision and F1 of the four heuristics on the test set.*

The three local indices are practically tied on this test set (ROC-AUC 0.9554-0.9557) and clearly beat Preferential Attachment (0.8490). The reason is visible in the data: 8.8% of the test positives but 99.6% of the random negatives have no common neighbour, so 'share at least one co-author' already separates most pairs. The F1-optimal thresholds learned on the training pairs (CN >= 1, Jaccard >= 0.0119, AA >= 0.2423) all amount to exactly that rule, which is why the three methods have identical precision (0.995), recall (0.912) and F1 (0.952). For PA the learned threshold is k(u)k(v) >= 10, giving F1 0.756.

6.2 ROC and Precision-Recall Curves

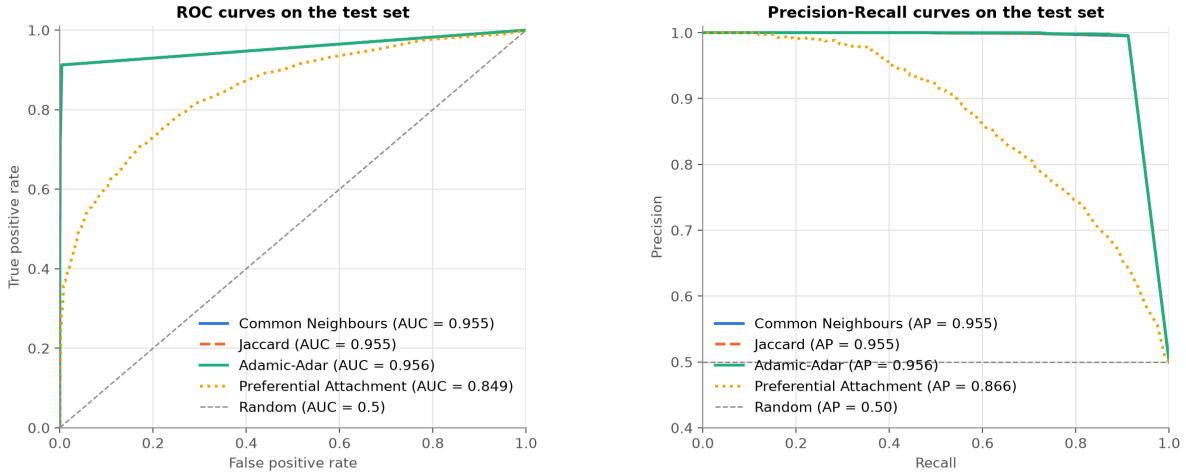


Figure 4: *ROC curves (left) and precision-recall curves (right) on the test set. CN, Jaccard and AA overlap almost perfectly; their curves bend at the share of positives with at least one common neighbour.*

6.3 Precision@K and a Harder Test with Hard Negatives

With random negatives Precision@K is 1.0 for every method up to K = 100 (Table 7): the top of every ranking is filled with true links, so this metric cannot separate the methods here. We therefore repeated the evaluation with the same positives against **hard negatives** - non-links that already share at least one co-author in the training graph.

Table 8: *Results against 2,684 hard negatives (distance-2 non-links).*

Method	ROC-AUC	AP	P@10	P@20	P@50	P@100
Common Neighbours	0.7776	0.8144	1.000	1.000	1.000	1.000
Jaccard	0.7594	0.7880	0.600	0.700	0.780	0.790
Adamic-Adar	0.8504	0.9004	1.000	1.000	1.000	1.000
Preferential Attachment	0.6438	0.6812	1.000	1.000	0.980	0.990

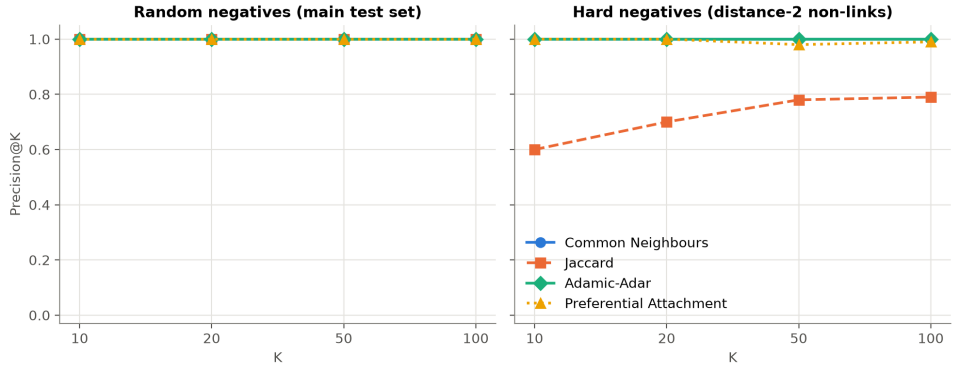


Figure 5: *Precision@K on the main test set (left) and with hard negatives (right). Common Neighbours is hidden under the Adamic-Adar line where both are 1.0.*

Now the methods separate: Adamic-Adar keeps the highest ROC-AUC (0.850) and AP (0.900), ahead of Common Neighbours (0.778), Jaccard (0.759) and PA (0.644). Jaccard's Precision@10 drops to 0.600: its top scores (1.0) go to small groups of low-degree authors with identical co-author lists, many of which never link.

6.4 Stability over Five Random Splits

Table 9: *Mean and standard deviation over seeds 42, 43, 44, 45, 46 (new split and new negatives each time).*

Method	ROC-AUC mean	ROC-AUC std	AP mean	AP std
Common Neighbours	0.9564	0.0029	0.9560	0.0028
Jaccard	0.9564	0.0028	0.9560	0.0023
Adamic-Adar	0.9568	0.0028	0.9569	0.0028
Preferential Attachment	0.8424	0.0043	0.8614	0.0034

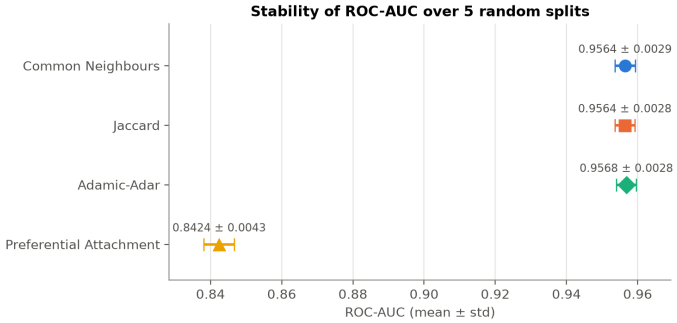


Figure 6: *ROC-AUC mean +/- standard deviation over five random splits.*

The standard deviations are below 0.004, so the ranking of the methods does not depend on one lucky split. The gap between the local indices and PA is about 0.114, much larger than the noise.

6.5 Comparison and Analysis: Why Adamic-Adar Wins

- **Clustering drives everything.** The graph is built from overlapping cliques (papers), so new links close triangles. All local indices capture this; PA does not look at shared neighbours at all.
- **Hubs are weak evidence.** The hubs of ca-GrQc are members of very large collaborations. Sharing such a hub tells little about a specific pair; AA discounts it with 1/log k(w), CN does not. This is exactly what the hard-negative test rewards.
- **Jaccard over-normalises.** Dividing by the union favours pairs of low-degree authors and penalises authors inside large groups, hurting its top-ranked predictions.
- **PA favours hubs.** PA scores pairs of popular authors highly even when they work in different groups. It still beats random because the network is degree-assortative, but it misses the locality of collaboration.

7. Recommendations

7.1 Candidate Generation and Filters

Scoring all 8,642,403 LCC pairs is unnecessary. For CN, Jaccard and AA every pair at distance 3 or more has no common neighbour and scores exactly 0, so it can never enter a Top-K list. Candidates are therefore the non-adjacent pairs at distance exactly 2: 63,288 pairs in the LCC. For PA this is a deliberate restriction that stops it from proposing hub pairs with no connection at all. Filters: no self-pairs, no pair already linked in the full graph, and no (u, v)/(v, u) duplicates (pairs are stored with u < v).

7.2 Validation: Do the Predictions Come True?

Table 10: *Hit rate@K: share of the Top-K training-graph predictions (out of 46,588 distance-2 candidates) that are held-out test links.*

K	Common Neighbours	Jaccard	Adamic-Adar	Preferential Attachment	Random
10	1.000	1.000	1.000	1.000	0.053
20	1.000	0.900	1.000	1.000	0.053
50	1.000	0.660	1.000	0.960	0.053
100	1.000	0.440	1.000	0.920	0.053
500	0.984	0.160	0.986	0.540	0.053
1000	0.894	0.487	0.918	0.377	0.053

91.2% of the test links are at distance 2 in the training graph, so a random distance-2 candidate is a real link with probability 0.053. Adamic-Adar and CN keep a hit rate of 1.0 up to K = 100 and 0.986 / 0.984 at K = 500; that is about 19 times better than random. Jaccard's hit rate falls to 0.440 at K = 100 and PA's to 0.540 at K = 500. A high hit rate shows that the top of the ranking is reliable. This is the property a recommender needs.

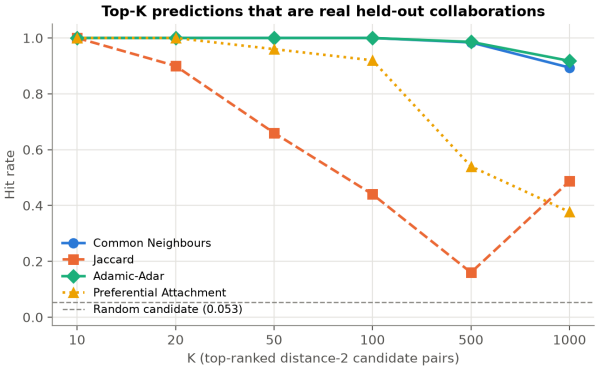


Figure 7: *Hit rate of the Top-K predictions made on the training graph.*

7.3 Top-20 Recommended Collaborations (Adamic-Adar)

Table 11: *Top-20 predicted new collaborations by Adamic-Adar on the full LCC (no pair is linked in the data).*

#	Author u	Author v	AA score	Common nbrs	Dist.	k(u)	k(v)	Shared co-authors (least-connected first)
1	45	46	10.598	42	2	49	42	2952 (deg 45), 4511 (deg 45), 6830 (deg 45), +39 more
2	4511	4513	10.592	42	2	45	43	2952 (deg 45), 6830 (deg 45), 8879 (deg 45), +39 more
3	4511	22691	10.592	42	2	45	77	2952 (deg 45), 6830 (deg 45), 8879 (deg 45), +39 more
4	46	4513	10.335	41	2	42	43	2952 (deg 45), 6830 (deg 45), 8879 (deg 45), +38 more
5	46	22691	10.335	41	2	42	77	2952 (deg 45), 6830 (deg 45), 8879 (deg 45), +38 more
6	5210	7444	6.699	25	2	33	28	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
7	7444	8727	6.699	25	2	28	33	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
8	7444	9869	6.699	25	2	28	41	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
9	7444	12710	6.699	25	2	28	33	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
10	7444	15245	6.699	25	2	28	33	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
11	7444	15933	6.699	25	2	28	41	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
12	7444	19446	6.699	25	2	28	33	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
13	7444	20576	6.699	25	2	28	41	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
14	7444	22504	6.699	25	2	28	33	15457 (deg 34), 23712 (deg 34), 20574 (deg 37), +22 more
15	2449	5210	6.392	24	2	34	33	9869 (deg 41), 15933 (deg 41), 20576 (deg 41), +21 more
16	2449	8727	6.392	24	2	34	33	9869 (deg 41), 15933 (deg 41), 20576 (deg 41), +21 more
17	2449	12710	6.392	24	2	34	33	9869 (deg 41), 15933 (deg 41), 20576 (deg 41), +21 more
18	2449	15245	6.392	24	2	34	33	9869 (deg 41), 15933 (deg 41), 20576 (deg 41), +21 more
19	2449	15457	6.392	24	2	34	34	9869 (deg 41), 15933 (deg 41), 20576 (deg 41), +21 more
20	2449	19446	6.392	24	2	34	33	9869 (deg 41), 15933 (deg 41), 20576 (deg 41), +21 more

All 20 pairs are at distance 2 and share between 24 and 42 co-authors; they involve only 17 distinct authors. For every pair, the group formed by the two authors and their shared co-authors has an edge density of 0.997-0.999: these are almost complete cliques, typical of very large multi-author papers or long-running consortia, in which the recommended pair is one of the few missing links. **Why recommended:** for example, authors 45 and 46 share 42 co-authors (degrees 49 and 42) but have never published together.

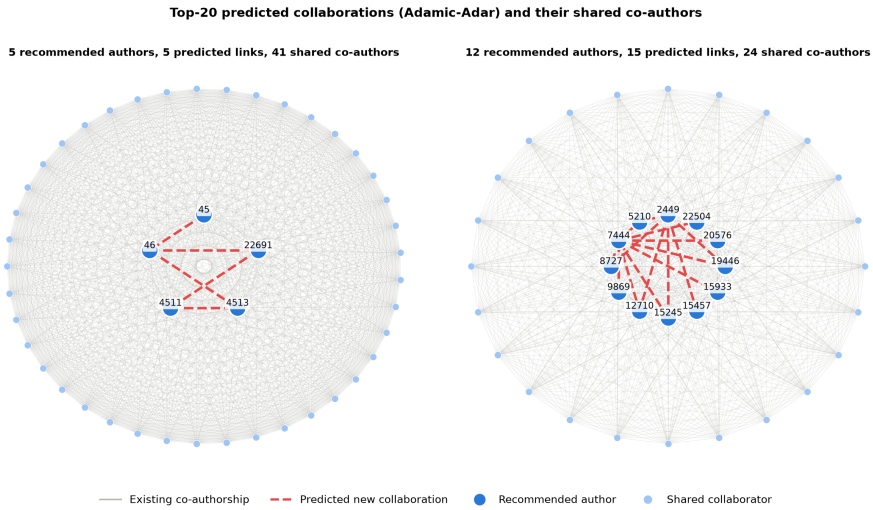


Figure 8: *The Top-20 recommendations: recommended authors (inner ring), their shared co-authors (outer ring), existing co-authorships (grey) and predicted collaborations (dashed red).*

7.4 Top-10 of the Other Methods

Table 12: *Top-10 recommendations of the other three heuristics (full table with explanations in outputs/top10_recommendations.csv).*

#	Common Neighbours: pair (score; common nbrs; k(u), k(v))	Jaccard: pair (score; common nbrs; k(u), k(v))	Preferential Attachment: pair (score; common nbrs; k(u), k(v))
1	45 - 46 (42; 42; 49, 42)	19012 - 20305 (1; 5; 5, 5)	13801 - 21012 (3645; 1; 45, 81)
2	4511 - 4513 (42; 42; 45, 43)	3595 - 18174 (1; 4; 4, 4)	13929 - 21281 (3555; 1; 45, 79)
3	4511 - 22691 (42; 42; 45, 77)	352 - 20533 (1; 3; 3, 3)	13801 - 21281 (3555; 1; 45, 79)
4	46 - 4513 (41; 41; 42, 43)	15666 - 15667 (1; 3; 3, 3)	18866 - 21281 (3476; 12; 44, 79)
5	46 - 22691 (41; 41; 42, 77)	6703 - 15596 (1; 3; 3, 3)	4511 - 22691 (3465; 42; 45, 77)
6	5210 - 7444 (25; 25; 33, 28)	5227 - 7015 (1; 3; 3, 3)	13929 - 22691 (3465; 1; 45, 77)
7	7444 - 8727 (25; 25; 28, 33)	195 - 19168 (1; 3; 3, 3)	12365 - 13801 (3465; 1; 77, 45)
8	7444 - 9869 (25; 25; 28, 41)	8730 - 13177 (1; 3; 3, 3)	13801 - 22691 (3465; 1; 45, 77)
9	7444 - 12710 (25; 25; 28, 33)	3411 - 4428 (1; 3; 3, 3)	46 - 22691 (3234; 41; 42, 77)
10	7444 - 15245 (25; 25; 28, 33)	3411 - 24734 (1; 3; 3, 3)	2654 - 21012 (3078; 1; 38, 81)

Common Neighbours shares 10 of its Top-10 pairs with Adamic-Adar, because both reward large overlaps. Jaccard shares none: its Top-10 pairs score 1.0 because the two authors have identical, small co-author lists. Preferential Attachment proposes pairs of hubs; many share only a single co-author, so these are 'popular' rather than 'close' recommendations.

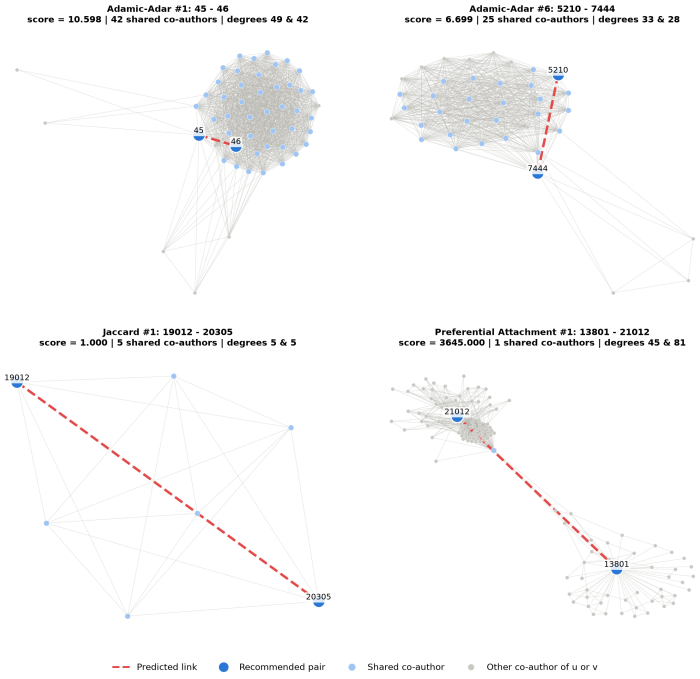


Figure 9: *Ego networks of the #1 pair of each method (plus the best pair from a second group for Adamic-Adar). Dark blue: recommended pair; light blue: shared co-authors; grey: other co-authors.*

8. Discussion

8.1 Interpretation of Results

Collaboration in this field is local and group-based. Once two authors share a collaborator, the probability that they collaborate is high, and the most informative shared collaborators are the less-connected ones. This is exactly the assumption behind Adamic-Adar, which is why it ranks best.

8.2 Strengths and Weaknesses

- **Strengths:** no training, fast (all test pairs are scored in well under a second), fully interpretable, and verified against NetworkX.
- **Weaknesses:** CN, JC and AA cannot score pairs at distance 3 or more (they are all 0), so they cannot recommend 'bridging' collaborations between groups. PA can, but with low precision. All four ignore time, paper counts and research topics.

8.3 Limitations and Threats to Validity

- **Random rather than temporal split:** without timestamps, 'future' links are simulated by hiding random edges. A temporal split (train on earlier years, test on later ones) would be more realistic.
- **Split bias:** test edges are drawn from non-spanning-tree edges, which lie on cycles and are therefore easier to predict than bridges. Results may be slightly optimistic.
- **Easy negatives:** random negatives are almost never at distance 2, which inflates ROC-AUC. The hard-negative results (ROC-AUC 0.644-0.850) are a more honest picture.
- **Balanced evaluation:** 1:1 sampling hides the real imbalance of about 643 non-links per link, so absolute precision values do not transfer to a deployed system. The hit-rate analysis over all distance-2 candidates is closer to deployment.
- **Data artefacts:** very large cliques dominate the top recommendations, and a missing link inside a clique may come from author-name disambiguation errors rather than a true missing collaboration. Author IDs are anonymised, so recommendations cannot be checked against real publications.

8.4 Challenges Encountered

Keeping the training graph connected (solved with the spanning-tree split), avoiding leakage when scoring training positives (solved by hiding the pair's own edge), the saturation of Precision@K with random negatives (solved with the hard-negative test and the hit-rate analysis), and the slow force-directed layout of the 4,158-node LCC (computed once with a fixed seed and cached).

9. Conclusion

9.1 Summary of the Work Done

We modeled the ca-GrQc co-authorship data as an undirected graph, characterised it (sparse, highly clustered, small-world, modular), and built a leakage-free link prediction pipeline. Four heuristics were implemented from scratch and compared. Adamic-Adar performed best (ROC-AUC 0.9568 ± 0.0028; hard-negative ROC-AUC 0.850) and was used to produce explained Top-20 collaboration recommendations, visualisations and a Cytoscape export.

9.2 Key Learnings and Insights

- Network structure alone predicts collaborations very well when the network is clustered.
- How the evaluation is designed (split, negatives, leakage) matters as much as the choice of algorithm: easy negatives make every local method look perfect.
- Down-weighting hubs (Adamic-Adar) is what separates the methods on hard cases.

9.3 Real-World Applicability

The pipeline can run as a 'researchers you may want to work with' feature: it is cheap, explainable ('you share 12 co-authors, e.g. ...') and can be recomputed nightly on a bibliographic database.

9.4 Future Improvements

- **Path-based scores** such as Katz [12] or Resource Allocation, which also reach pairs at distance 3 or more.
- **Node embeddings** (Node2Vec [11]) or graph neural networks to learn structural features.
- **Supervised classifier** (e.g. logistic regression or gradient boosting) trained on CN, JC, AA, PA and community features.
- **Temporal evaluation** on time-stamped arXiv data, and **weighted edges** (number of joint papers).
- **Diversity-aware re-ranking** so that one large clique does not dominate the recommendation list.

10. References

[1] D. Liben-Nowell and J. Kleinberg, "The link-prediction problem for social networks," *Journal of the American Society for Information Science and Technology*, 58(7), pp. 1019-1031, 2007.

[2] J. Leskovec, J. Kleinberg and C. Faloutsos, "Graph evolution: Densification and shrinking diameters," *ACM Transactions on Knowledge Discovery from Data*, 1(1), 2007.

[3] L. A. Adamic and E. Adar, "Friends and neighbors on the Web," *Social Networks*, 25(3), pp. 211-230, 2003.

[4] M. E. J. Newman, "Clustering and preferential attachment in growing networks," *Physical Review E*, 64, 025102, 2001.

[5] A.-L. Barabasi and R. Albert, "Emergence of scaling in random networks," *Science*, 286(5439), pp. 509-512, 1999.

[6] P. Jaccard, "Etude comparative de la distribution florale dans une portion des Alpes et du Jura," *Bulletin de la Societe Vaudoise des Sciences Naturelles*, 37, pp. 547-579, 1901.

[7] L. Lu and T. Zhou, "Link prediction in complex networks: A survey," *Physica A*, 390(6), pp. 1150-1170, 2011.

[8] V. D. Blondel, J.-L. Guillaume, R. Lambiotte and E. Lefebvre, "Fast unfolding of communities in large networks," *Journal of Statistical Mechanics*, P10008, 2008.

[9] A. Hagberg, D. Schult and P. Swart, "Exploring network structure, dynamics, and function using NetworkX," *Proc. 7th Python in Science Conference (SciPy)*, pp. 11-15, 2008.

[10] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, 12, pp. 2825-2830, 2011.

[11] A. Grover and J. Leskovec, "node2vec: Scalable feature learning for networks," *Proc. ACM KDD*, pp. 855-864, 2016.

[12] L. Katz, "A new status index derived from sociometric analysis," *Psychometrika*, 18(1), pp. 39-43, 1953.

[13] P. Shannon et al., "Cytoscape: A software environment for integrated models of biomolecular interaction networks," *Genome Research*, 13(11), pp. 2498-2504, 2003.

[14] J. Leskovec and A. Krevl, "SNAP Datasets: Stanford Large Network Dataset Collection," 2014. https://snap.stanford.edu/data/ca-GrQc.html

[15] M. E. J. Newman, "The structure of scientific collaboration networks," *Proceedings of the National Academy of Sciences*, 98(2), pp. 404-409, 2001.